

Phase 1 - Week 4 - Lesson Day 4

A/B Testing + Mini Project 1 (Phase 1 Wrap-Up)

Today's Goal

Today we learn A/B testing - a simple, real-world way companies use hypothesis testing to make decisions. Then we finish Phase 1 with a mini project that uses everything you learned in Weeks 1 to 4.

- Understand what A/B testing is and why companies use it
- Run a simple A/B test
- Finish Mini Project 1 - a full EDA plus A/B test on one dataset

1. What Is A/B Testing?

Simple definition: A/B testing means showing two different versions of something to two different groups of people, then checking which version performs better. "A" is usually the current version, and "B" is the new version being tested.

Example: a company wants to know if a green "Buy Now" button gets more clicks than a blue one. They show the green button to half their visitors (Group A) and the blue button to the other half (Group B). Then they compare the click rates using the same hypothesis testing tools from Day 2.

A/B testing is really just a hypothesis test, applied to a business decision. The null hypothesis is "both versions perform the same." The alternative is "one version performs better."

2. Running a Simple A/B Test

```
from scipy import stats

# Click rates: 1 = clicked, 0 = did not click
group_a_clicks = [1, 0, 0, 1, 0, 0, 1, 0, 0, 0] # old button (10 visitors)
group_b_clicks = [1, 1, 0, 1, 1, 0, 1, 1, 0, 1] # new button (10 visitors)

rate_a = sum(group_a_clicks) / len(group_a_clicks)
rate_b = sum(group_b_clicks) / len(group_b_clicks)

print("Group A click rate:", rate_a)
print("Group B click rate:", rate_b)

t_stat, p_value = stats.ttest_ind(group_a_clicks, group_b_clicks)
print("p-value:", p_value)

if p_value < 0.05:
    print("The difference is likely REAL - the new button performs differently")
else:
    print("Not enough proof yet - could be random chance")
```

Notice this uses the exact same t-test from Day 2 - A/B testing isn't a brand new tool, it's just hypothesis testing applied directly to a business question.

Mini Project 1: Full EDA + A/B Test

This is the big combined project for the end of Phase 1. It uses skills from every week so far: Pandas (Week 2), visualization and statistics (Week 3), and hypothesis testing (Week 4).

Open **day-4-ab-testing-mini-project.ipynb**. It contains a complete, solved example mini project on a simulated dataset - use it as your template.

Mini Project Steps

1. Load a dataset (your own from Week 2, or the simulated one in the notebook).
2. Do a short EDA: check for missing values, look at `.describe()`, plot at least one chart.
3. Pick two groups inside your data to compare (for example, two categories, or two time periods).
4. Write a null hypothesis and an alternative hypothesis for your comparison, in plain words.
5. Run a t-test comparing the two groups.
6. Write your conclusion: is the difference likely real, or could it be random chance? What would you recommend based on this result?

This mini project is a simple but complete version of what a real data analyst does at work: look at data, ask a clear question, test it properly, and explain the result in plain words that anyone can understand - not just other data scientists.

Once you and Halimo finish the mini project, Phase 1 is complete! Next up is Phase 2: Machine Learning Core, starting with Week 5 - ML Foundations.